

Heaven's Light Is Our Guide

Rajshahi University of Engineering & Technology



Lab Report 01

Course Title: Digital Techniques Sessional

Course No: ECE 2112

Report Title: Digital Logic Gates & Circuits

Submitted By	Submitted To
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Introduction

This report presents a collection of fundamental digital logic circuits implemented and simulated to demonstrate core concepts of digital electronics, including universal gate derivations (NAND/NOR), a Full Adder, and an 8-Bit Binary to BCD converter. All circuits were designed and simulated using Logisim.

Task 1: All Fundamental Logic Gates using NOR Gate

The NOR gate is a universal gate, meaning any other logic gate can be constructed using only NOR gates. This circuit demonstrates three separate derivations — NOT, OR, and AND — built entirely from NOR gates.

Description

- **NOT Gate:** Formed by connecting both inputs of a single NOR gate together, so it acts as an inverter.
- **OR Gate:** Formed by feeding two NOR gates in cascade — the first NOR gate combines A and B, and the second NOR gate (used as a NOT) inverts that result back to a plain OR.
- **AND Gate:** Formed using De Morgan's construction — each input is first inverted by its own NOR-based NOT gate, and the inverted signals are then combined by a final NOR gate to produce the AND output.

Truth Table 1 — NOT Gate (from NOR)

A	Output (NOT A)
0	1
1	0

Truth Table 2 — OR Gate (from NOR)

A	B	Output (A OR B)
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table 3 — AND Gate (from NOR)

A	B	Output (A AND B)
0	0	0
0	1	0
1	0	0
1	1	1

Circuit Diagram

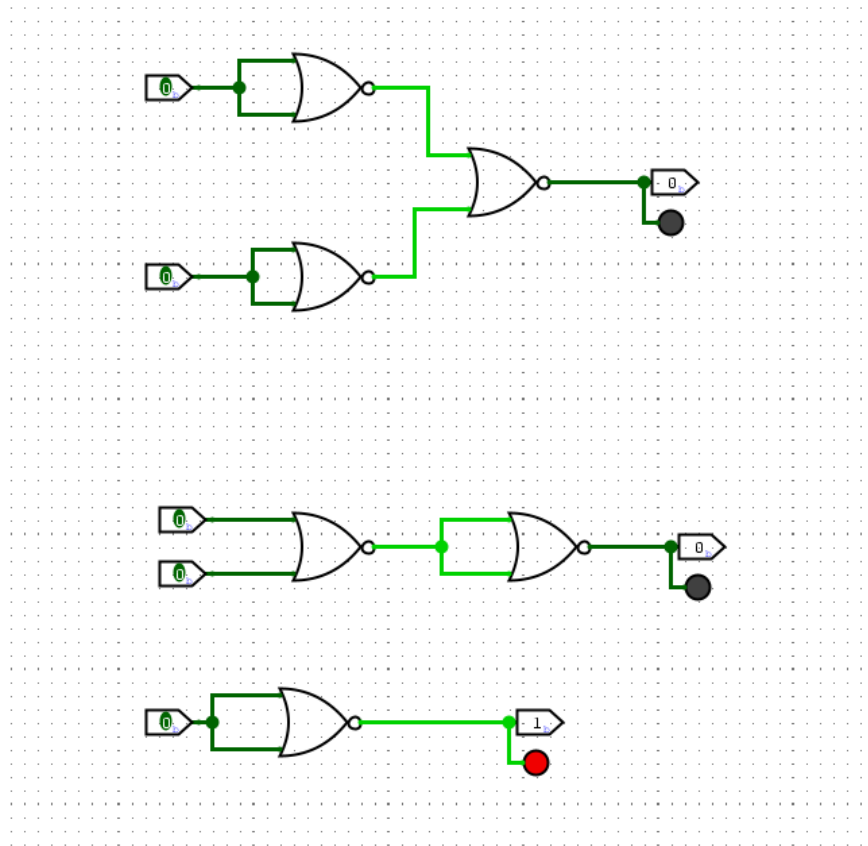


Figure 1: All Fundamental Logic Gates using NOR Gate

Task 2: All Fundamental Logic Gates using NAND Gate

The NAND gate is also a universal gate. Similar to the NOR-based approach, this circuit demonstrates three separate derivations — AND, OR, and NOT — built entirely from NAND gates.

Description

- **AND Gate:** Formed by cascading two NAND gates — the first NAND combines A and B, and the second NAND (used as a NOT) inverts that result back to a plain AND.
- **OR Gate:** Formed using De Morgan's construction — each input is first inverted by its own NAND-based NOT gate, and the inverted signals are then combined by a final NAND gate to produce the OR output.
- **NOT Gate:** Formed by connecting both inputs of a single NAND gate together, so it acts as an inverter.

Truth Table 1 — AND Gate (from NAND)

A	B	Output (A AND B)
0	0	0
0	1	0
1	0	0
1	1	1

Truth Table 2 — OR Gate (from NAND)

A	B	Output (A OR B)
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table 3 — NOT Gate (from NAND)

A	Output (NOT A)
0	1
1	0

Circuit Diagram

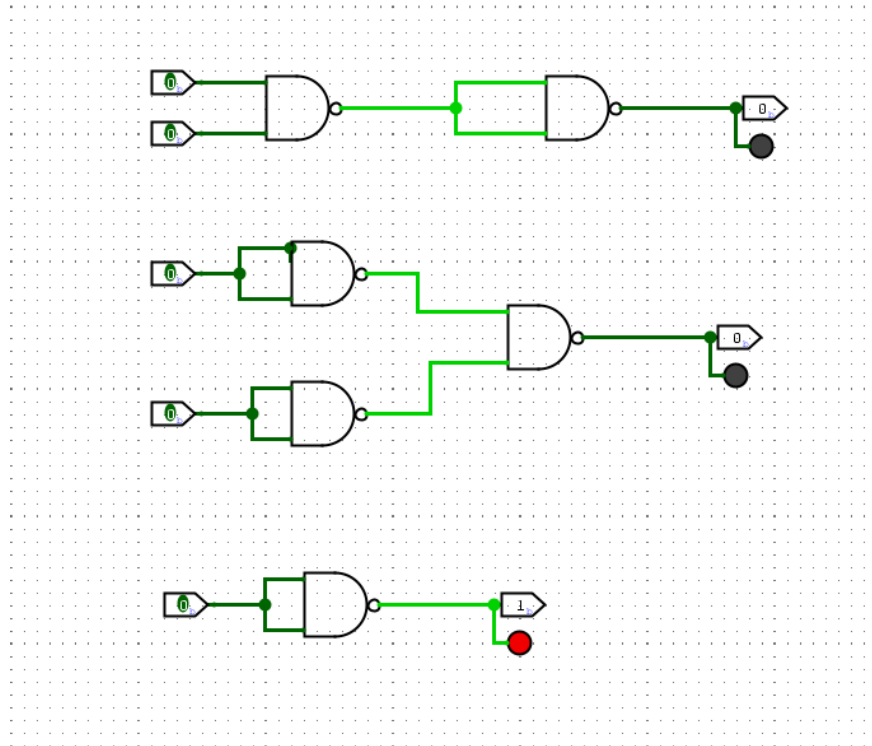


Figure 2: All Fundamental Logic Gates using NAND Gate

Task 3: Full Adder

A Full Adder is a combinational circuit that performs the addition of three binary bits: two significant bits (A and B) and a carry-in bit (Cin) from a previous addition stage. It produces two outputs: the Sum and the Carry-out (Cout). The circuit is implemented in two ways: using a built-in Adder component (with Cin/Cout), and using two cascaded Half Adders built from XOR/AND/OR gates.

Description

- **Built-in Full Adder block:** Takes A, B, and Cin as inputs, and directly outputs Sum and Cout.
- **Half Adder 1:** Takes inputs A and B, and produces an intermediate Sum (A XOR B) and an intermediate Carry (A AND B).
- **Half Adder 2 + OR stage:** Takes the intermediate Sum from Half Adder 1 along with Cin to produce the final Sum, while the two intermediate carries are combined with an OR gate to produce the final Cout.

Truth Table 1 — Half Adder 1 (A, B)

A	B	Sum1 (A XOR B)	Carry1 (A AND B)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth Table 2 — Half Adder 2 (Sum1, Cin)

Sum1	Cin	Sum (Sum1 XOR Cin)	Carry2 (Sum1 AND Cin)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth Table 3 — Final Full Adder (A, B, Cin)

A	B	Cin	Sum	Cout (Carry1 OR Carry2)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Circuit Diagram

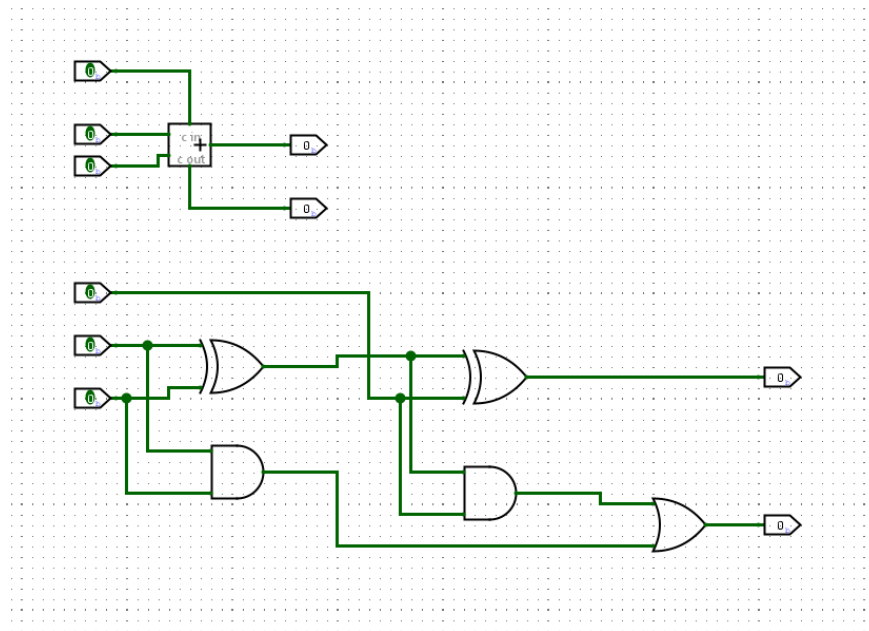


Figure 3: Full Adder

Task 4: 8-Bit Binary to BCD Converter

This circuit converts an 8-bit binary input into its equivalent BCD (Binary Coded Decimal) representation and displays it on three 7-segment displays (Hundreds, Tens, and Units place).

Description

- Takes an 8-bit binary number as input (range: 0–255).
- Converts the binary value into three BCD digits: Hundreds (100s), Tens (10s), and Units (1s).
- Each BCD digit drives its own 7-segment display, showing the decimal equivalent of the binary input.
- Commonly implemented using the Double Dabble (Shift-and-Add-3) algorithm to convert binary to BCD.
- Useful in digital systems where binary data needs to be shown in human-readable decimal form, such as counters, digital clocks, and measurement displays.

Truth Table (Sample Binary to BCD Conversions)

Binary Input	Decimal	Hundreds	Tens	Units
00000000	0	0	0	0
00001001	9	0	0	9
00011001	25	0	2	5
01100100	100	1	0	0
11111111	255	2	5	5

Circuit Diagram

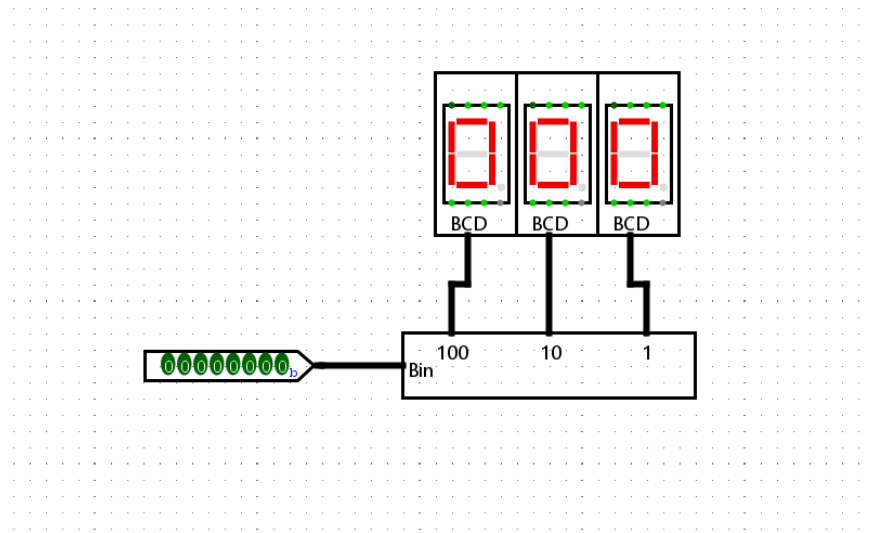


Figure 4: 8-Bit Binary to BCD Converter

Tools Used

- Logic circuit simulation software (Logisim)